

Human Activity Recognition Using Deep Learning Models on the KU-HAR Dataset

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Abstract—Human Activity Recognition (HAR) has become a new field of active research due to its applications in healthcare, sports monitoring, assisted living, and intelligent mobile systems. Recent advances in deep learning have significantly improved the ability to automatically recognize human activities from wearable devices and smartphone sensors. This work presents a comparative evaluation of three deep learning architectures for HAR using the publicly available KU-HAR dataset, which contains recordings from smartphone accelerometers and gyroscopes sensors acquired from 90 participants performing 18 different activities.

Three representative architectures were implemented and evaluated: a one-dimensional Convolutional Neural Network (CNN1D), a Bidirectional Long Short-Term Memory network (BiLSTM), and a hybrid Convolutional Recurrent Neural Network (CRNN). All models were trained under the same preprocessing pipeline and evaluated using Accuracy, Balanced Accuracy, Macro F1-score, Weighted F1-score, model complexity, and computational cost.

Experimental results show that the BiLSTM achieved the highest recognition performance, reaching a test accuracy of 97.37% and a Macro F1-score of 98.08%. However, this improvement required considerably longer training times. In contrast, the CNN1D presented the lowest computational cost while maintaining acceptable performance, whereas the CRNN offered a compromise between accuracy and computational efficiency. These findings highlight the trade-off between predictive performance and computational requirements when selecting deep learning models for smartphone-based HAR applications.

Index Terms—CNN, Wearable, movement, smartphones.

I. INTRODUCTION

HUMAN Activity Recognition (HAR) has emerged as one of the most important research areas in artificial intelligence due to its wide range of applications in healthcare, rehabilitation, sports performance analysis, elderly assistance, human-computer interaction [?], [?], and smart environments. The fast expansion of smartphones equipped with inertial sensors has enabled continuous monitoring of human movements without requiring specialized hardware, making smartphone-based HAR an attractive and cost-effective solution.

Traditional machine learning approaches for HAR rely on handcrafted feature extraction followed by conventional classifiers. Although these methods have demonstrated satisfactory performance in controlled scenarios, they often

require significant domain expertise and may not generalize well to complex movement patterns. Deep learning techniques overcome these limitations by automatically learning hierarchical representations directly from raw sensor signals, eliminating the need for manual feature engineering.

Among deep learning approaches, Convolutional Neural Networks (CNNs) have shown excellent capability for extracting local temporal patterns from inertial signals, while Long Short-Term Memory (LSTM) networks are particularly effective at modeling temporal dependencies present in sequential data. More recently, hybrid Convolutional Recurrent Neural Networks (CRNNs) have combined the strengths of both architectures by integrating convolutional feature extraction with recurrent temporal modeling.

Despite the increasing popularity of these models, selecting the most appropriate architecture remains a challenge because prediction accuracy must be balanced against computational cost and training time, especially for real-time applications deployed on mobile or embedded devices.

This work presents a comparative evaluation of three representative deep learning architectures for smartphone-based HAR using the KU-HAR dataset. The proposed project try to investigate the performance of a one-dimensional Convolutional Neural Network (CNN1D), a Bidirectional Long Short-Term Memory network (BiLSTM), and a Convolutional Recurrent Neural Network (CRNN). The comparison considers not only classification performance but also computational efficiency, including training time and model complexity, providing practical insights into the strengths and limitations of each architecture.

II. RELATED WORK

HAR has been extensively investigated over the past decade due to the widespread availability of wearable devices and smartphones equipped with inertial sensors. Early HAR systems primarily relied on handcrafted feature extraction techniques combined with traditional machine learning classifiers such as Support Vector Machines (SVM), Decision Trees (DT), k-Nearest Neighbors (k-NN), and Random Forests (RF). Although these methods achieved satisfactory performance under controlled conditions, they required significant domain knowledge for feature engineering and often exhibited limited generalization capabilities.

The emergence of deep learning has transformed HAR by enabling end-to-end learning directly from raw sensor

signals. Convolutional Neural Networks (CNNs) have demonstrated remarkable capability for automatically extracting local temporal features from multivariate time-series data while reducing the need for manual feature design. Several studies have reported that one-dimensional CNNs provide an efficient solution for smartphone-based HAR due to their relatively low computational complexity and parallel processing capabilities.

Recurrent neural networks, particularly Long Short-Term Memory (LSTM) architectures, have also become popular for HAR because they explicitly model temporal dependencies present in sequential sensor data. Bidirectional LSTM (BiLSTM) networks further improve temporal representation by simultaneously processing sequences in both forward and backward directions, allowing the model to capture contextual information from the complete activity sequence.

More recently, hybrid Convolutional Recurrent Neural Networks (CRNNs) have combined convolutional feature extraction with recurrent temporal modeling. The convolutional layers learn local motion patterns while the recurrent layers capture long-term temporal dependencies, often leading to improved recognition accuracy. However, these architectures usually require higher computational resources and longer training times than simpler CNN-based models.

Although numerous studies have reported competitive performance using these deep learning approaches, selecting the most appropriate architecture remains application-dependent. In particular, practical deployments on mobile and embedded platforms require balancing classification accuracy against computational cost, memory requirements, and training efficiency. Motivated by these considerations, this work presents a comparative evaluation of CNN1D, BiLSTM, and CRNN architectures using the KU-HAR dataset under identical preprocessing and evaluation protocols.

III. MATERIALS AND METHODS

This section describes the dataset, preprocessing pipeline, exploratory data analysis, deep learning architectures, and experimental configuration employed throughout the study. All experiments were conducted under identical conditions to ensure a fair comparison among the evaluated models.

A. KU-HAR Dataset

The experiments were conducted using the publicly available KU-HAR (KUST Human Activity Recognition) dataset, which was specifically designed for smartphone-based human activity recognition. The dataset contains inertial measurements collected from 90 participants (75 male and 15 female) performing 18 daily activities while carrying a smartphone equipped with a tri-axial accelerometer and gyroscope.

The original dataset consists of 1,945 raw recordings that were subsequently segmented into 20,750 non-overlapping

subsamples. Each subsample represents a three-second activity window sampled at 100 Hz, resulting in 300 temporal observations for each sensor channel.

Each sample contains six synchronized channels corresponding to the three accelerometer axes (X, Y, and Z) and the three gyroscope axes (X, Y, and Z). Consequently, each observation is represented by a multivariate time series of dimensions 6×300 , yielding 1,800 numerical features. Additionally, each record includes the activity label, window length, and sample identifier.

The dataset comprises eighteen activity classes, including static activities (Stand, Sit, Lay), transitional movements (Stand-Sit and Lay-Stand), locomotion activities (Walk, Run, Stair-Up, Stair-Down), and more complex dynamic actions such as Jump, Push-up, Sit-up, Pick, and Table Tennis.

B. Exploratory Data Analysis

Before model development, an exploratory data analysis was conducted to verify the integrity of the dataset, examine the distribution of activity classes, and identify relevant temporal patterns in the accelerometer and gyroscope signals. This analysis was also used to evaluate whether the sensor measurements contained sufficient discriminative information for the subsequent Human Activity Recognition experiments.

1) *Dataset Integrity*: The original KU-HAR dataset contained 20,750 activity samples and 1,803 columns. Of these columns, 1,800 corresponded to inertial sensor measurements, while the remaining three represented the activity label, sequence length, and subject identifier. Each sample contained 300 temporal observations obtained from six channels: three accelerometer axes and three gyroscope axes.

The initial inspection confirmed that the dataset contained no missing or infinite values. In addition, all activity samples were represented by sequences of 300 time steps, allowing the sensor data to be consistently reshaped into tensors of dimensions 6×300 . This uniform representation was subsequently used as the input format for all deep learning architectures evaluated in this study.

2) *Class Distribution*: The distribution of samples among the eighteen activity categories was examined to identify possible class imbalance. As shown in Fig. ??, the number of observations varied considerably across classes. While several activities contained more than 1,700 samples, some categories were represented by fewer than 500 observations.

The ratio between the most represented and least represented classes was approximately 8.41. This imbalance motivated the use of class-weighted cross-entropy during model training and the inclusion of Balanced Accuracy and Macro F1-score as evaluation metrics.

3) *Signal Magnitude Analysis*: To provide an orientation-independent representation of body motion, the magnitude of the accelerometer and gyroscope signals was computed

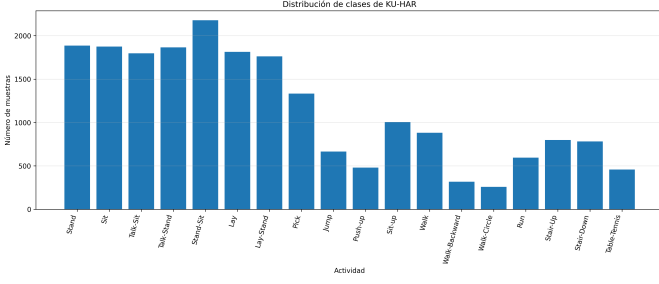


Figure 1. Distribution of samples across the eighteen activity classes in the KU-HAR dataset.

from the three spatial axes. The resulting signals summarize the overall intensity of linear acceleration and angular velocity throughout each activity window.

Figures ?? and ?? present representative magnitude signals extracted from the accelerometer and gyroscope, respectively. Compared with the individual sensor axes, the magnitude representation facilitates the visualization of movement intensity while reducing the influence of smartphone orientation.

Dynamic activities exhibit larger oscillations and greater signal variability, whereas static activities produce considerably more stable magnitude profiles. These observations suggest that the magnitude signals contain meaningful information that can support activity discrimination.



Figure 2. Representative accelerometer magnitude signals computed from the three spatial axes.

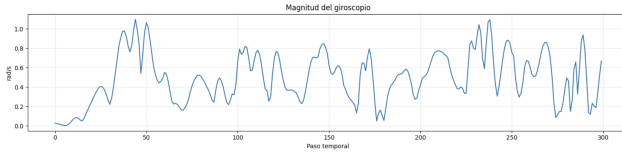


Figure 3. Representative gyroscope magnitude signals computed from the three spatial axes.

4) *Comparison of Activity Magnitudes:* To further investigate the separability of the recorded activities, representative magnitude signals from multiple activity classes were compared.

Static activities such as standing and sitting exhibit nearly constant magnitude profiles, while transitional movements generate localized fluctuations associated with changes in body posture. In contrast, dynamic activities including walking, running, and jumping display periodic oscillations with considerably larger amplitudes. These differences indicate that movement intensity and temporal behavior vary substantially across activity categories.

Although some activities present partially overlapping magnitude profiles, the complete six-channel sensor representation preserves additional spatial information that can be exploited by deep learning architectures to improve classification performance.

5) *Summary of the Exploratory Analysis:* The exploratory analysis confirmed the suitability of the KU-HAR dataset for Human Activity Recognition. The dataset exhibited consistent temporal windows without missing values and a manageable degree of class imbalance. Moreover, the magnitude visualizations revealed distinctive motion patterns among static, transitional, and dynamic activities, providing qualitative evidence that the inertial signals contain sufficient discriminative information for deep learning-based classification.

C. Data Preprocessing

To ensure a fair comparison among the evaluated models, a unified preprocessing pipeline was applied to all experiments.

The complete dataset was first divided into training (70%), validation (15%), and testing (15%) subsets using stratified sampling to preserve the original class distribution.

Instead of conventional standardization, robust normalization was adopted to reduce the influence of extreme observations. Median and interquartile range (IQR) statistics were computed exclusively from the training subset to avoid information leakage. Each sensor channel was independently normalized according to

$$x' = \frac{x - \text{Median}}{\text{IQR}}, \quad (1)$$

followed by clipping the normalized values to the interval $[-10, 10]$.

This strategy improves numerical stability during optimization while preserving the temporal structure of the multivariate signals. Finally, all samples were converted into tensors with dimensions 6×300 , maintaining the original synchronization among the six sensor channels.

D. Deep Learning Architectures

Three deep learning models were implemented to evaluate different strategies for learning temporal patterns from multivariate inertial sensor data. All architectures received the same input representation, consisting of six synchronized sensor channels and 300 temporal samples per activity window. The models were trained and evaluated under identical experimental conditions to ensure a fair comparison.

The evaluated architectures include a one-dimensional Convolutional Neural Network (CNN1D), a Bidirectional Long Short-Term Memory network (BiLSTM), and a hybrid Convolutional Recurrent Neural Network (CRNN). Each model represents a different approach to feature

extraction and temporal modeling, allowing the analysis of their strengths and computational requirements.

1) *One-Dimensional Convolutional Neural Network*: The CNN1D model was designed to automatically learn local temporal patterns from the inertial signals through successive convolutional operations. The network consists of multiple one-dimensional convolutional layers followed by batch normalization, ReLU activation functions, max-pooling layers, and dropout regularization. The extracted feature maps are flattened and passed through fully connected layers to perform the final activity classification.

CNNs are particularly effective for HAR because neighboring samples in inertial signals often contain correlated motion information. Furthermore, one-dimensional convolutions significantly reduce computational complexity compared to recurrent architectures while maintaining competitive recognition performance.

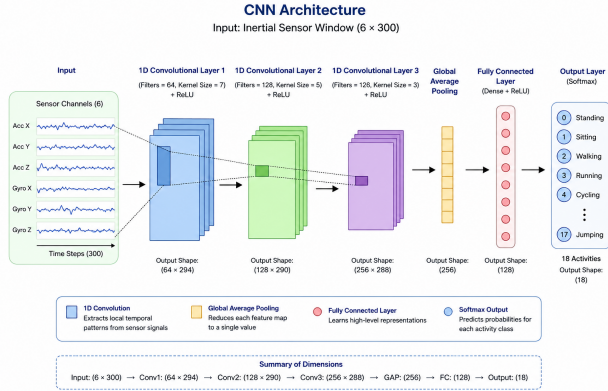


Figure 4. Architecture of the proposed CNN1D model.

E. Experimental Setup

All experiments were implemented in Python using PyTorch and PyTorch Lightning. Training was performed using the Adam optimizer with categorical cross-entropy loss. Early stopping and model checkpointing were employed to prevent overfitting and preserve the best-performing model during training.

The dataset was divided into training (70%), validation (15%), and testing (15%) subsets using stratified sampling. All models were trained using identical data partitions, ensuring a fair comparison among the evaluated architectures.

Model performance was assessed using multiple evaluation metrics, including Accuracy, Balanced Accuracy, Macro F1-score, Weighted F1-score, confusion matrices, and training time. Additionally, the number of trainable parameters was recorded to evaluate model complexity.

IV. EXPERIMENTAL RESULTS

This section presents the experimental results obtained by the CNN1D, BiLSTM, and CRNN architectures on the KU-HAR test subset. The models were evaluated using

Table I
TRAINING HYPERPARAMETERS

Parameter	Value
Optimizer	Adam
Learning Rate	0.001
Loss Function	Cross Entropy
Batch Size	64
Epochs	50
Early Stopping	Yes
Random Seed	42

Table II
PERFORMANCE OF THE CNN1D MODEL

Metric	Result
Accuracy	82.78%
Balanced Accuracy	88.86%
Macro F1-score	89.29%
Weighted F1-score	83.38%
Trainable Parameters	260,754
Training Time	181 s

Accuracy, Balanced Accuracy, Macro F1-score, Weighted F1-score, number of trainable parameters, and training time. Learning curves and confusion matrices were also analyzed to assess convergence behavior and class-level performance.

A. CNN1D Results

The CNN1D model achieved a test accuracy of 82.78%, a Balanced Accuracy of 88.86%, a Macro F1-score of 89.29%, and a Weighted F1-score of 83.38%. The difference between the standard accuracy and the Balanced Accuracy indicates that the model performed better when each activity class was considered with equal importance.

The learning curves presented in Fig. 5 show a stable reduction in both training and validation loss. Similarly, the accuracy curves improved progressively throughout training. The relatively small separation between the training and validation curves suggests that the regularization mechanisms were effective in limiting overfitting.

The CNN1D contained 260,754 trainable parameters and required approximately 181 seconds to complete training. Therefore, it represented the computationally least expensive architecture evaluated in this study.

B. BiLSTM Results

The BiLSTM obtained the highest predictive performance among the evaluated architectures, achieving a test accuracy of 97.37% and a Macro F1-score of 98.08%. These results demonstrate the effectiveness of bidirectional recurrent processing for modeling the temporal dependencies contained in the inertial signals.

As shown in Fig. 7, the training and validation curves followed similar trajectories during optimization. The validation loss remained stable while the validation accuracy approached the training accuracy, indicating strong generalization and limited evidence of overfitting.

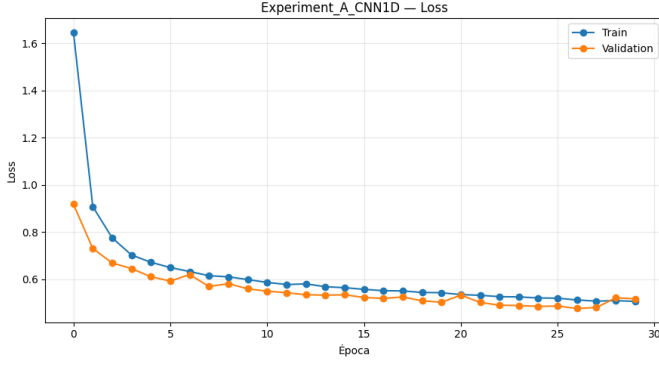


Figure 5. Training and validation learning curves of the CNN1D model.

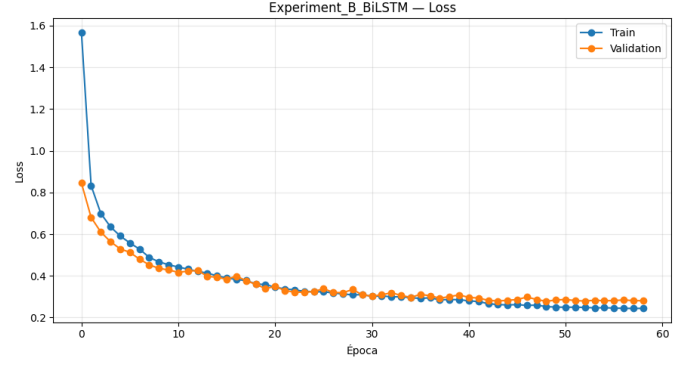


Figure 7. Training and validation learning curves of the BiLSTM model.

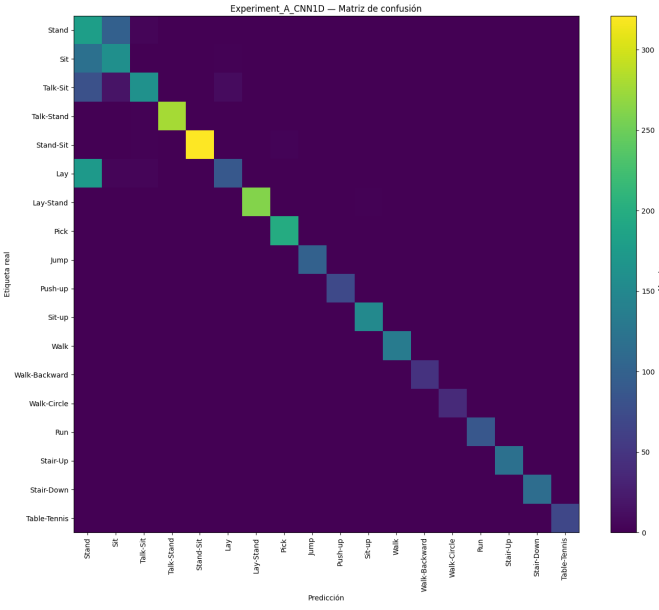


Figure 6. Confusion matrix obtained by the CNN1D model on the test subset.

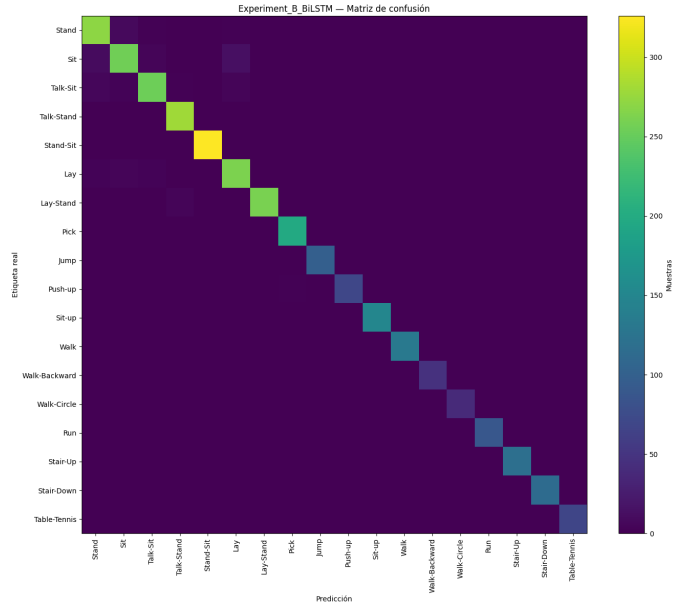


Figure 8. Confusion matrix obtained by the BiLSTM model on the test subset.

Table III
PERFORMANCE OF THE BiLSTM MODEL

Metric	Result
Accuracy	97.37%
Macro F1-score	98.08%
Test Loss	0.074
Training Time	≈1.5 h

Despite its superior classification performance, the BiLSTM required approximately 1.5 hours of training. This was substantially longer than the training times recorded for the CNN1D and CRNN models. The result highlights the increased computational cost associated with recurrent processing of complete temporal sequences.

The confusion matrix presented in Fig. 8 confirms that the BiLSTM achieved consistent recognition across the eighteen activity classes. Only a limited number of samples were assigned to incorrect categories, indicating that bidirectional sequence modeling was effective in distinguishing both static and dynamic activities.

C. CRNN Results

The CRNN achieved a test accuracy of 85.93%, a Balanced Accuracy of 90.76%, a Macro F1-score of 91.00%, and a Weighted F1-score of 86.03%. These results represent an improvement over the CNN1D in all reported classification metrics.

The learning curves in Fig. 9 demonstrate stable optimization behavior. Both training and validation performance improved progressively, although the final results remained below those obtained by the BiLSTM.

The CRNN contained 361,106 trainable parameters and required approximately seven minutes to complete training. Therefore, its computational cost was higher than that of the CNN1D but substantially lower than that of the BiLSTM.

As observed in Fig. ??, the hybrid model correctly classified most samples while reducing several of the errors observed in the CNN1D. Nevertheless, some confusion remained

Table IV
PERFORMANCE OF THE CRNN MODEL

Metric	Result
Accuracy	85.93%
Balanced Accuracy	90.76%
Macro F1-score	91.00%
Weighted F1-score	86.03%
Trainable Parameters	361,106
Training Time	≈7 min

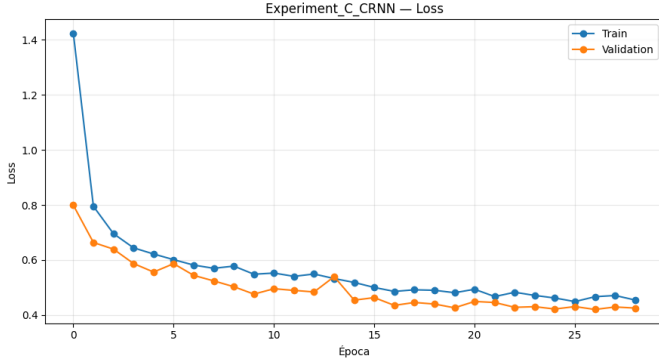


Figure 9. Training and validation learning curves of the CRNN model.

among activities with related motion structures, suggesting that the convolutional compression stage may have removed part of the temporal information required for more precise discrimination.

D. Comparative Analysis

Table V summarizes the principal results obtained by the three evaluated architectures. The BiLSTM achieved the highest test accuracy and Macro F1-score, substantially outperforming both convolution-based alternatives. The CRNN obtained the second-highest classification performance, while the CNN1D produced the lowest accuracy but required the shortest training time and the smallest number of trainable parameters.

The difference between Accuracy and Macro F1-score was particularly relevant for the CNN1D and CRNN models. Because the KU-HAR dataset presents an unequal number of samples across activity classes, conventional accuracy is influenced more strongly by the most represented categories. Macro F1-score assigns the same importance to every class and therefore provides a complementary evaluation of performance under class imbalance.

From a computational perspective, the CNN1D required approximately 181 seconds to train, whereas the CRNN required approximately seven minutes. In contrast, the BiLSTM required approximately 1.5 hours, making it nearly 30 times slower than the CNN1D and approximately 13 times slower than the CRNN.

Figure 10 illustrates the difference in classification performance among the evaluated models. The BiLSTM exceeded the CNN1D by 14.59 percentage points in accuracy and exceeded the CRNN by 11.44 percentage points.

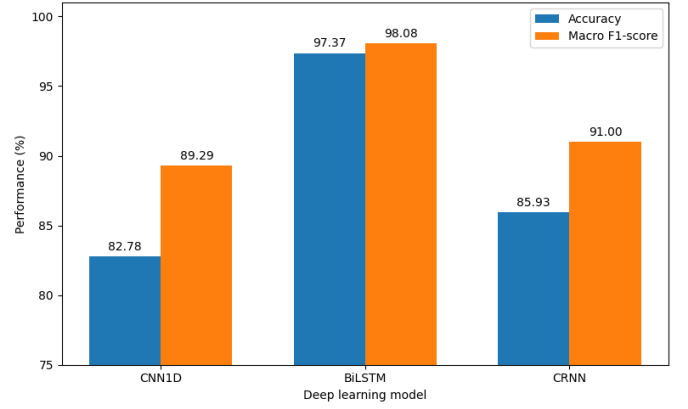


Figure 10. Comparison of test Accuracy and Macro F1-score among the evaluated models.

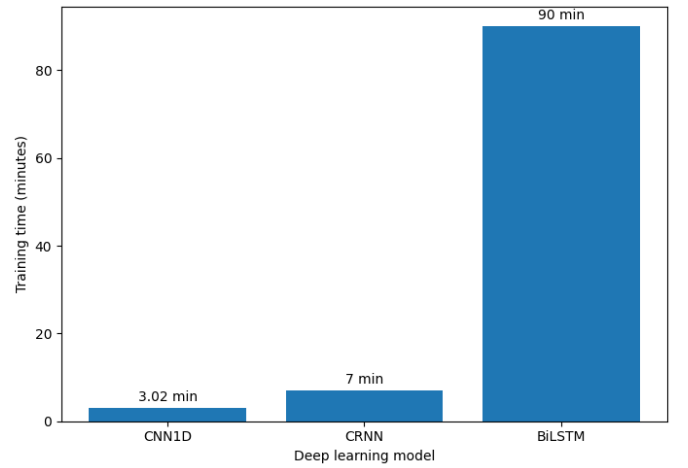


Figure 11. Comparison of the approximate training time required by each architecture.

As shown in Fig. 11, the superior predictive performance of the BiLSTM was associated with a considerably higher computational cost. The CNN1D was the fastest architecture, while the CRNN provided an intermediate training time.

V. DISCUSSION

The experimental results demonstrate that the three evaluated architectures exhibit distinct strengths depending on the balance between classification performance and computational efficiency. Although all models were capable of learning meaningful representations from the KU-HAR dataset, significant differences were observed in their ability to model temporal dependencies and their associated computational requirements.

A. Performance Analysis

Among the evaluated architectures, the BiLSTM consistently achieved the highest recognition performance, reaching a test accuracy of 97.37% and a Macro F1-score of 98.08%. These results indicate that explicitly modeling temporal dependencies in both forward and backward

Table V
COMPARATIVE PERFORMANCE OF THE EVALUATED DEEP LEARNING MODELS

Model	Accuracy	Balanced Accuracy	Macro F1	Weighted F1	Parameters	Training Time
CNN1D	82.78%	88.86%	89.29%	83.38%	260,754	181 s
BiLSTM	97.37%	—	98.08%	—	—	≈1.5 h
CRNN	85.93%	90.76%	91.00%	86.03%	361,106	≈7 min

directions provides a significant advantage for recognizing complex human activities from inertial sensor signals.

Unlike convolutional models, recurrent architectures preserve the temporal order of the complete sequence during feature extraction. This capability is particularly beneficial for HAR because many activities are characterized not only by individual sensor values but also by the evolution of movement over time. Consequently, the BiLSTM was able to distinguish activities with similar instantaneous motion patterns but different temporal dynamics more effectively than the other evaluated models.

B. CNN and CRNN Behavior

Although the CRNN combined convolutional feature extraction with recurrent processing, its performance remained below that of the BiLSTM. A possible explanation is that the initial convolutional layers compressed part of the temporal information before it reached the recurrent component. While convolution efficiently extracts local motion features, some long-range temporal dependencies may be partially attenuated during pooling operations.

The KU-HAR dataset contains relatively clean and well-segmented activity windows. Under these conditions, directly processing the complete temporal sequence through a BiLSTM appears to preserve richer contextual information than first applying convolutional compression. Consequently, the additional convolutional stage did not provide sufficient complementary information to compensate for the loss of temporal resolution.

C. Computational Efficiency

The CNN1D obtained the lowest classification accuracy but also required the shortest training time and the smallest number of trainable parameters. This result confirms one of the principal advantages of convolutional architectures: efficient feature extraction with relatively low computational complexity.

Despite its lower predictive performance, the CNN1D achieved an accuracy exceeding 82%, demonstrating that local temporal patterns alone contain sufficient information to correctly recognize most activities. Therefore, CNN-based architectures remain attractive candidates for resource-constrained environments such as smartphones, wearable devices, and embedded systems, where computational efficiency is often a primary design consideration.

D. Trade-off Between Accuracy and Computational Cost

An important outcome of this study is the evident trade-off between recognition performance and computational

Table VI
SUMMARY OF THE STRENGTHS AND LIMITATIONS OF THE EVALUATED MODELS

Property	CNN1D	BiLSTM	CRNN
Recognition Accuracy	Medium	High	Medium-High
Training Time	Low	High	Medium
Model Complexity	Low	High	Medium
Temporal Modeling	Limited	Excellent	Good
Mobile Deployment	Excellent	Limited	Good

cost. The BiLSTM provided the highest classification accuracy but required approximately 1.5 hours of training, making it nearly thirty times slower than the CNN1D and considerably slower than the CRNN.

Conversely, the CNN1D completed training in approximately three minutes while maintaining competitive performance. The CRNN occupied an intermediate position, requiring approximately seven minutes of training while providing moderate improvements over the CNN1D.

These findings suggest that the optimal architecture depends on the intended application. Systems prioritizing maximum recognition accuracy may benefit from recurrent architectures such as BiLSTM, whereas real-time or embedded applications may favor CNN-based models due to their substantially lower computational requirements.

E. Limitations

Although the proposed comparison provides valuable insights into the performance of representative deep learning architectures, several limitations should be acknowledged. First, all experiments were conducted using a single publicly available dataset. Consequently, the observed performance may vary when evaluated on datasets collected under different acquisition protocols or sensor configurations.

Second, only three representative architectures were investigated. More recent sequence models based on attention mechanisms or transformer architectures may provide additional improvements in recognition performance while reducing computational complexity.

Finally, inference time and energy consumption were not evaluated in this study. These aspects are particularly relevant for real-time smartphone applications and should be considered in future research.

As summarized in Table VI, each architecture presents different strengths depending on the requirements of the target application.

VI. CONCLUSION

This work presented a comparative evaluation of three deep learning architectures for smartphone-based Human Activity Recognition using the KU-HAR dataset. A one-dimensional Convolutional Neural Network (CNN1D), a Bidirectional Long Short-Term Memory network (BiLSTM), and a hybrid Convolutional Recurrent Neural Network (CRNN) were implemented under identical pre-processing, training, and evaluation conditions to ensure a fair comparison.

Among the evaluated models, the BiLSTM achieved the highest predictive performance, reaching a test accuracy of 97.37% and a Macro F1-score of 98.08%. These results demonstrate the importance of explicitly modeling long-term temporal dependencies when classifying multivariate inertial sensor signals. In contrast, the CNN1D provided the shortest training time and the lowest computational complexity, making it a suitable alternative for applications where computational resources are limited. The CRNN offered an intermediate solution, improving classification performance over the CNN1D while maintaining a substantially lower computational cost than the BiLSTM.

Overall, the experimental results confirm that selecting an appropriate architecture for Human Activity Recognition requires balancing predictive performance and computational efficiency according to the target application. While recurrent architectures maximize recognition accuracy, convolutional approaches remain attractive for real-time and embedded implementations due to their lower computational requirements.

Future research will focus on evaluating more recent sequence-learning architectures, including Transformer-based and attention-based models, to determine whether higher recognition accuracy can be achieved without substantially increasing computational cost. In addition, future studies should investigate cross-subject generalization, real-time inference performance, and energy consumption on mobile and embedded devices to better assess the practical deployment of Human Activity Recognition systems.

REFERENCES

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